

The Slime Science Mystery

Name: _____

Date: _____

Year 3 English — Reading

ACARA v9.0: AC9E3LY05

Learning Intention

I am learning to answer literal and inferential questions about an informative text.

Success Criteria

- I can find facts stated directly in the text.
- I can answer questions using clues from the text.
- I can explain a science process in my own words.

The Slime Science Mystery

Slime has become one of the most popular things kids love to make at home. But have you ever wondered what actually makes slime so stretchy, gooey and fun to squish?

Slime is a simple example of a polymer. A polymer is a substance made of long chains of molecules linked together, a bit like a chain made of thousands of tiny paperclips hooked together. Normally, glue is a runny liquid because its molecule chains can slide past each other easily.

When you add a special ingredient — usually a mixture containing borax or a saline solution — something interesting happens. This ingredient links the long glue molecule chains together at different points, like adding extra clips connecting the paperclip chain in random spots. This process is called cross-linking.

Once cross-linking happens, the mixture can no longer flow like a liquid, but it isn't a solid either. It becomes a strange in-between substance, stretchy and slow-moving, which scientists call a "non-Newtonian fluid." That's the secret behind slime's stretchy, gooey texture!

Scientists use the study of polymers for far more than just slime. Polymers are used to make plastics, rubber, and even some medical materials. So next time you're stretching a piece of homemade slime, remember — you're actually holding real chemistry in your hands.

Part A — Comprehension Questions

Q1. What is slime an example of? _____

Q2. What ingredient is usually added to glue to make slime? _____

Q3. True or False: Slime is a normal liquid. _____

Q4. What is the process called when molecule chains link together?

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Part B — More Comprehension Questions

Q5. Why is glue normally runny, according to the text?

Q6. Name 2 other things polymers are used to make (besides slime).

1. _____ 2. _____

Q7. What term do scientists use to describe slime's strange in-between texture?

Part C — Challenge: Think Deeper

A student wants to explain to a younger sibling why slime is stretchy, using the paperclip chain comparison from the text.

Q8. Explain, in your own words, how the paperclip chain comparison helps explain cross-linking.

Q9. The text says making slime is 'real chemistry.' Explain what this means using evidence from the text.

Q10. Predict what might happen if you added too much of the cross-linking ingredient to the glue. Use the text to support your reasoning.

Answer Key — Part A & B

A1: A polymer.

A2: Borax or a saline solution.

A3: False — slime is neither a normal liquid nor a solid; it is a strange in-between substance.

A4: Cross-linking.

A5: Because its molecule chains can slide past each other easily.

A6: Plastics and rubber (accept medical materials).

A7: A non-Newtonian fluid.

Answer Key — Part C

C8: Just like clipping extra paperclips randomly along a chain makes it harder for the chain to move freely, cross-linking connects glue's molecule chains at extra points, making the mixture thicker and stretchier instead of runny.

C9: It means that making slime involves real scientific processes (polymers, molecules, cross-linking) — not just a fun craft activity, but an actual chemical reaction happening in the mixture.

C10: Any reasonable prediction (e.g. the slime might become too stiff, hard or rubbery, since more cross-linking would connect even more of the molecule chains together, reducing how much it can stretch and flow) — teacher to check reasoning is text-supported.